



SBD HEALTHCARE PVT.LTD.

GIDC ESTATE, PANOLI-394116

MATERIAL SAFETY DATA SHEET

Product : 2-Bromo-2-nitro-1,3-propanediol

SECTION 1 - CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

MSDS Name : 2-Bromo-2-nitro-1,3-propanediol

CAS # : 52-51-7

Company Identification : SBD Healthcare Pvt. Ltd. Panoli

For Information Call : +91-9429428251

SECTION 2 - COMPOSITION, INFORMATION ON INGREDIENTS

Chemical Name : 2-Bromo-2-nitro-1,3-propanediol

Purity : NLT 98.0 %

CAS # : 52-51-7

SECTION 3 - HAZARDS IDENTIFICATION.

Appearance: slightly yellow crystals.

Warning! Toxic. Flammable solid. Eye contact may result in permanent eye damage. Causes respiratory tract irritation. Harmful if swallowed. Causes eye and skin irritation. May be harmful if absorbed through the skin. May cause central nervous system depression. Hygroscopic (absorbs moisture from the air). Dangerous for the environment.

Target Organs: Central nervous system.

Eye: Contact may cause severe eye irritation and possible eye damage. May cause chemical conjunctivitis and corneal damage.

Skin: May be harmful if absorbed through the skin. May cause irritation and dermatitis. May cause cyanosis of the extremities.

Ingestion: Harmful if swallowed. May cause gastrointestinal irritation with nausea, vomiting and diarrhea. Ingestion of large amounts may cause CNS depression.

Inhalation: Causes respiratory tract irritation. Aspiration may lead to pulmonary edema. Vapors may cause dizziness or suffocation. Causes irritation of the mucous membrane and upper respiratory tract. May cause burning sensation in the chest.

Chronic: Not available.



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SECTION 4 - FIRST AID MEASURES**Description of first-aid measures**

Eyes: Immediately flush eyes with plenty of water for at least 15 minutes, occasionally lifting the upper and lower eyelids. Get medical aid immediately.

Skin: Get medical aid. Flush skin with plenty of water for at least 15 minutes while removing contaminated clothing and shoes. Wash clothing before reuse.

Ingestion: Never give anything by mouth to an unconscious person. Get medical aid immediately. Do NOT induce vomiting. If conscious and alert, rinse mouth and drink 2-4 cupfuls of milk or water.

Inhalation: Get medical aid immediately. Remove from exposure and move to fresh air immediately. If breathing is difficult, give oxygen. If breathing has ceased apply artificial respiration using oxygen and a suitable mechanical device such as a bag and a mask.

Notes to Physician: Treat symptomatically and supportively.

SECTION 5 - FIRE FIGHTING MEASURES**Extinguishing media**

General Information: As in any fire, wear a self-contained breathing apparatus in pressure-demand, MSHA/NIOSH (approved or equivalent), and full protective gear. Vapors may form an explosive mixture with air. Vapors can travel to a source of ignition and flash back. Will burn if involved in a fire. Use water spray to keep fire-exposed containers cool. Containers may explode in the heat of a fire. Flammable solid.

Extinguishing Media: For small fires, use dry chemical, carbon dioxide, water spray or alcohol-resistant foam. For large fires, use water spray, fog, or alcohol-resistant foam. In case of fire, use water, dry chemical, chemical foam, or alcohol-resistant foam. Use water spray to cool fire-exposed containers. Water may be ineffective. Do NOT use straight streams of water.

Flash Point: 130 deg C (266.00 deg F)

Autoignition Temperature: Not available.

Explosion Limits, Lower:Not available.

Upper: Not available.

NFPA Rating: (estimated) Health: ; Flammability: ; Instability:



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SECTION 6 - ACCIDENTAL RELEASE MEASURES

General Information: Use proper personal protective equipment as indicated in Section 8.

Spills/Leaks: Absorb spill with inert material (e.g. vermiculite, sand or earth), then place in suitable container. Sweep up, then place into a suitable container for disposal. Remove all sources of ignition. Use a spark-proof tool. A vapor suppressing foam may be used to reduce vapors.

SECTION 7 - HANDLING AND STORAGE

Handling: Ground and bond containers when transferring material. Use spark-proof tools and explosion proof equipment. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep container tightly closed. Keep away from heat, sparks and flame. Do not pressurize, cut, weld, braze, solder, drill, grind, or expose empty containers to heat, sparks or open flames.

Storage: Keep away from heat, sparks, and flame. Keep away from sources of ignition. Store in a cool, dry, well-ventilated area away from incompatible substances. Flammables-area. Store protected from moisture.

SECTION 8 - EXPOSURE CONTROLS, PERSONAL PROTECTION.

Engineering Controls: Use adequate general or local explosion-proof ventilation to keep airborne levels to acceptable levels.

OSHA Vacated PELs: 2-Bromo-2-nitro-1,3-propanediol: No OSHA Vacated PELs are listed for this chemical.

Personal Protective Equipment

Eyes: Wear safety glasses and chemical goggles if splashing is possible.

Skin: Wear appropriate protective gloves and clothing to prevent skin exposure.

Clothing: Wear appropriate protective clothing to minimize contact with skin.

Respirators: Wear a NIOSH/MSHA or European Standard EN 149 approved full-facepiece airline respirator in the positive pressure mode with emergency escape provisions.



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SECTION 9 - PHYSICAL AND CHEMICAL PROPERTIES**Information on basic physical and chemical properties**

- A) Physical State:** Crystals
- B) Appearance:** slightly yellow
- C) Odor:** None reported.
- D) pH:** Not available.
- E) Vapor Pressure:** Not available.
- F) Vapor Density:** Not available.
- G) Evaporation Rate:** Not available.
- H) Viscosity:** Not available.
- I) Boiling Point:** Not available.
- J) Freezing/Melting Point:** > 120 deg C
- K) Decomposition Temperature:** 130 deg C
- L) Solubility:** 25g/100ml (22 c)
- M) Specific Gravity/Density:** Not available.
- N) Molecular Formula:** $C_3H_6BrNO_4$
- O) Molecular Weight:** 199.99

SECTION 10 -STABILITY AND REACTIVITY

Chemical Stability: Stable under normal temperatures and pressures.

Conditions to Avoid: Ignition sources, excess heat, exposure to moist air or water.

Incompatibilities with Other Materials: Strong oxidizing agents, strong reducing agents, strong bases, acid chlorides, acid anhydrides, moisture.

Hazardous Decomposition Products: Nitrogen oxides, carbon monoxide, carbon dioxide, hydrogen bromide, nitrogen gas.

Hazardous Polymerization: Has not been reported.



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SECTION 11 - TOXICOLOGICAL INFORMATION**Information on toxicological effects****LD50/LC50:****CAS# 52-51-7:**

Draize test, rabbit, eye: 5 mg;

Draize test, rabbit, skin: 500 mg/24H Mild;

Draize test, rabbit, skin: 80 mg Moderate;

Inhalation, rat: LC50 = >5 gm/m³/6H;Inhalation, rat: LC50 = 800 mg/m³/4H;

Oral, mouse: LD50 = 270 mg/kg;

Oral, mouse: LD50 = 194 mg/kg;

Oral, rabbit: LD50 = 190 mg/kg;

Oral, rat: LD50 = 180 mg/kg;

Oral, rat: LD50 = 267 mg/kg;

Skin, rat: LD50 = 1600 mg/kg;

Carcinogenicity:

CAS# 52-51-7: Not listed by ACGIH, IARC, NTP, or CA Prop 65.

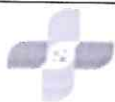
SECTION 12 - ECOLOGICAL INFORMATION

Ecotoxicity: Bacteria: Phytobacterium phosphoreum: EC50 = 0.41-0.90 mg/L; 5,15,30 minutes;
Microtox test; 15 degrees C

SECTION 13 - DISPOSAL CONSIDERATIONS

Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. US EPA guidelines for the classification determination are listed in 40 CFR Parts 261.3. Additionally, waste generators must consult state and local hazardous waste regulations to ensure complete and accurate classification.

RCRA P-Series: None listed.**RCRA U-Series:** None listed



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SECTION 14 - TRANSPORT INFORMATION**UN Number**

DOT (US): N/A IATA: N/A IMDG: N/A

UN Proper Shipping Name

DOT (US)/IATA:

Not dangerous goods

IMDG/ARD/RID:

Not dangerous goods

Transport Hazard Class(es)

DOT (US): N/A IATA: N/A IMDG: N/A

Packing Group

DOT (US): N/A IATA: N/A IMDG: N/A

Environmental Hazards

DOT (US): None IATA: None IMDG: None

Special Precautions for User

None

SECTION 15 – REGULATORY INFORMATION

This safety data sheet complies with the requirements of WHMIS (Canada), OSHA 1910.1200 (US), and EU Regulation

EC No. 1907/2006 (European Union).

Safety, Health and Environmental Regulations/Legislation Specific for the Substance or Mixture**A) Canada**

DSL/NDSL Status: This product is not listed on the Canadian DSL/NDSL.

B) United States

TSCA Status: This product is not listed on the US EPA TSCA.

Chemical Safety Assessment

S 26 In case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

S 37/39 Wear suitable gloves and eye/face protection.



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S 61 Avoid release to the environment. Refer to special instructions
/safety data sheets.

SECTION 16 - ADDITIONAL INFORMATION

The information above is believed to be accurate and represents the best information and references currently available to us. However, we make no warranty of merchantability or any other warranty, express or implied, with respect to such information, and we assume no liability resulting from its use. User should make their own investigations to determine the suitability of the information for their purposes. In no way shall SBD Healthcare Pvt. Ltd. Panoli be liable for any claims, losses, or damages of any third party or for lost profit or any special, indirect, incidental, consequential, or exemplary damages, how so ever arising, even if SBD Healthcare Pvt. Ltd. Panoli has been advised of the possibility of such damages.

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